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Turbine engine including a water system

Abstract

A turbine engine includes a spool with a compressor for providing a compressed air flow to a turbine, a shaft that connects the compressor and a turbine such that the compressor and the turbine rotate together, and a fuel supply. A combustor located downstream of the compressor receives and combusts the compressed air flow and the fuel supply, generating combustion gases. The combustion gases rotate the turbine, in turn rotating the spool. A balance piston for applying an axial force on the spool includes a rotating boundary portion and a fixed enclosure portion that does not rotate, defining a balance piston cavity. A water system provides water to the balance piston cavity, heat transfer from the balance piston to the water causes expansion of the water, the water in the balance piston cavity being balance piston fluid that is pressurized and applies an axial force to the spool.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure generally relates to a turbine including a water system.

BACKGROUND

(2) A turbine engine generally includes a fan and a core section arranged in flow communication with one another. A combustor is arranged in the core section to generate combustion gases for driving a turbine in the core section of the turbine engine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Features and advantages of the present disclosure will be apparent from the following description of various exemplary embodiments, as illustrated in the accompanying drawings, wherein like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.
- (2) FIG. 1 is a schematic cross-sectional view of a turbine engine with a water system, taken along a longitudinal centerline axis of the turbine engine, according to an embodiment of the disclosure.
- (3) FIG. 2 is a schematic view of a turbine engine with multiple thrust cavities, according to an embodiment of the disclosure.
- (4) FIG. 3 is a detailed schematic view of multiple thrust cavities, according to an embodiment of the disclosure.
- (5) FIG. 4 is a detailed schematic view of multiple thrust cavities, according to an embodiment of the disclosure.
- (6) FIG. 5 is a detailed schematic view of a thrust cavity, according to an embodiment of the disclosure.
- (7) FIG. 6 is a detailed schematic view of multiple thrust cavities, according to an embodiment of the disclosure.
- (8) FIG. 7 is a detailed schematic view of multiple thrust cavities, according to an embodiment of the disclosure.
- (9) FIG. 8 is a schematic view of a balance piston system, according to an embodiment of the disclosure.
- (10) FIG. 9 is a schematic view of a balance piston system, according to an embodiment of the disclosure.
- (11) FIG. 10 is a schematic view of a balance piston system, according to an embodiment of the disclosure.

DETAILED DESCRIPTION

- (12) Features, advantages, and embodiments of the present disclosure are set forth or apparent from a consideration of the following detailed description, drawings, and claims. Moreover, the following detailed description is exemplary and intended to provide further explanation without limiting the scope of the disclosure as claimed.
- (13) Various embodiments are discussed in detail below. While specific embodiments are discussed, this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without departing from the present disclosure.
- (14) As used herein, the terms “first” and “second” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.
- (15) The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.
- (16) The terms “forward” and “aft” refer to relative positions within a turbine engine or vehicle, and refer to the normal operational attitude of the turbine engine or vehicle. For example, with regard to a turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.
- (17) The terms “coupled,” “fixed,” “attached,” and the like, refer to both direct coupling, fixing, attaching, or connecting, as well as indirect coupling, fixing, attaching, or connecting through one or more intermediate components or features, unless otherwise specified herein.
- (18) As used herein, the terms “low,” “mid” (or “mid-level”), and “high,” or their respective comparative degrees (e.g., “lower” and “higher”, where applicable), when used with compressor,

turbine, shaft, fan, or turbine engine components, each refers to relative pressures, relative speeds, relative temperatures, and/or relative power outputs within an engine unless otherwise specified. For example, a “low-power” setting defines the engine configured to operate at a power output lower than a “high-power” setting of the engine, and a “mid-level power” setting defines the engine configured to operate at a power output higher than a “low-power” setting and lower than a “high-power” setting. The terms “low,” “mid” (or “mid-level”) or “high” in such aforementioned terms may additionally, or alternatively, be understood as relative to minimum allowable speeds, pressures, or temperatures, or minimum or maximum allowable speeds, pressures, or temperatures relative to normal, desired, steady state, etc., operation of the engine.

(19) The various power levels of the turbine engine detailed herein are defined as a percentage of a sea level static (SLS) maximum engine rated thrust. Low-power operation includes, for example, less than thirty percent (30%) of the SLS maximum engine rated thrust of the turbine engine. Mid-level power operation includes, for example, thirty percent (30%) to eighty-five (85%) of the SLS maximum engine rated thrust of the turbine engine. High-power operation includes, for example, greater than eighty-five percent (85%) of the SLS maximum engine rated thrust of the turbine engine. The values of the thrust for each of the low-power operation, the mid-level power operation, and the high-power operation of the turbine engine are exemplary only, and other values of the thrust can be used to define the low-power operation, the mid-level power operation, and the high-power operation.

(20) Here and throughout the specification and claims, range limitations are combined, and interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

(21) The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

(22) As used herein, the terms “axial” and “axially” refer to directions and orientations that extend substantially parallel to a centerline of the turbine engine. Moreover, the terms “radial” and “radially” refer to directions and orientations that extend substantially perpendicular to the centerline of the turbine engine. In addition, as used herein, the terms “circumferential” and “circumferentially” refer to directions and orientations that extend arcuately about the centerline of the turbine engine.

(23) Turbine engines incorporate thrust pistons in order to axially support the spools. Use of thrust pistons reduces the load on mechanical thrust bearings. Thrust pistons use compressed gas, such as air, to support a portion of the overall axial load on the respective spool. Reducing the load on the thrust bearings provides a longer service life for thrust bearings. Turbine engines additionally incorporate balance pistons, which function similar to thrust pistons, allowing for variability in the amount of axial load applied. Water as a piston fluid allows for easier pumping while in a liquid state, and provides heat transfer away from internal components in changing phase from liquid water to steam.

(24) As used herein, the term “interface” can refer alternatively to one or more seals, bearings, or defined gaps between components, where one component is movable (e.g., rotating) relative to the other. A seal may be any one of an oil seal, a piston seal, a carbon seal, a labyrinth seal, or any other configuration of a seal or a material of a seal as may be conceived to facilitate the movement of one component relative to another component while retaining pressure in the respective cavity. A bearing may be any one of a rolling element bearing, such as a single row ball bearing, a double row ball bearing, a roller bearing, a needle bearing, a tapered roller bearing, or any other rolling element bearing as may be conceived to facilitate the movement of one component relative to another component while retaining pressure in the respective cavity. A bearing may also be a hydrodynamic bearing, a friction bearing, or a sliding bearing, or any other similarly non-rolling-element bearing as may be conceived to facilitate the movement of one component relative to

another component while retaining pressure in the respective cavity. A defined gap may be of any size and of any shape as may be conceived to facilitate the movement of one component relative to another component while retaining pressure in the respective cavity, and as may be required to result in an acceptable leak rate through the interface, where appropriate. Additionally, the interface may conceivably include any combination of types of seals, types of bearings and sizes and shapes of gaps. Additionally, interface types may be applied in any combination. For example, bearings and seals are commonly applied together and such a combination may also be applied to the interfaces discussed herein.

(25) Referring now to the drawings, FIG. 1 is a schematic cross-sectional view of a turbine engine **10**, including a water system **92**, taken along a longitudinal centerline axis **12** (provided for reference) of the turbine engine **10**, according to an embodiment of the present disclosure. As shown in FIG. 1, the turbine engine **10** has an axial direction A (extending parallel to the longitudinal centerline axis **12**) and a radial direction R that is normal to the axial direction A. In general, the turbine engine **10** includes a fan section **14** and a turbo-engine **16** disposed downstream from the fan section **14**.

(26) The turbo-engine **16** includes an outer casing **18** that is substantially tubular and defines an annular core inlet **20**. As schematically shown in FIG. 1, the outer casing **18** encases, in serial flow relationship, a compressor section **21** including a booster or a low-pressure compressor (LPC) **22** followed downstream by a high-pressure compressor (HPC) **24**, a combustor **26**, a turbine section **27**, including a high-pressure turbine (HPT) **28**, followed downstream by a low-pressure turbine (LPT) **30**, and one or more core exhaust nozzles **32**. A high-pressure (HP) shaft **34** drivingly connects the HPT **28** to the HPC **24** to rotate the HPT **28** and the HPC **24** in unison. The HPT **28** is drivingly coupled to the HP shaft **34** to rotate the HP shaft **34** when the HPT **28** rotates. A low-pressure (LP) shaft **36** drivingly connects the LPT **30** to the LPC **22** to rotate the LPT **30** and the LPC **22** in unison. The LPT **30** is drivingly coupled to the LP shaft **36** to rotate the LP shaft **36** when the LPT **30** rotates. The compressor section **21**, the combustor **26**, the turbine section **27**, and the one or more core exhaust nozzles **32** together define a core air flow path **33**.

(27) For the embodiment depicted in FIG. 1, the fan section **14** includes a fan **38** (e.g., a variable pitch fan) having a plurality of fan blades **40** coupled to a disk **42** in a spaced apart manner. As depicted in FIG. 1, the fan blades **40** extend outwardly from the disk **42** generally along the radial direction R. Each fan blade **40** is rotatable relative to the disk **42** about a pitch axis P by virtue of the fan blades **40** being operatively coupled to an actuator **44** configured to collectively vary the pitch of the fan blades **40** in unison. The fan blades **40**, the disk **42**, and the actuator **44** are together rotatable about the longitudinal centerline axis **12** via a fan shaft **45** that is powered by the LP shaft **36** across a power gearbox, also referred to as a gearbox assembly **46**. The gearbox assembly **46** is shown schematically in FIG. 1. The gearbox assembly **46** includes a plurality of gears for adjusting the rotational speed of the fan shaft **45** and, thus, the fan **38** relative to the LP shaft **36**.

(28) Referring still to the exemplary embodiment of FIG. 1, the disk **42** is covered by a rotatable fan hub **48** aerodynamically contoured to promote an airflow through the plurality of fan blades **40**. In addition, the fan section **14** includes an annular fan casing or a nacelle **50** that circumferentially surrounds the fan **38**, at least a portion of the turbo-engine **16**, or both. The nacelle **50** is supported relative to the turbo-engine **16** by a plurality of circumferentially spaced outlet guide vanes **52**. Moreover, a downstream section **54** of the nacelle **50** extends over an outer portion of the turbo-engine **16** to define a bypass airflow passage **56** therebetween. The one or more core exhaust nozzles **32** may extend through the nacelle **50** and be formed therein. In this example, the one or more core exhaust nozzles **32** include one or more discrete nozzles that are spaced circumferentially about the nacelle **50**. Other arrangements of the core exhaust nozzles **32** may be used including, for example, a single core exhaust nozzle that is annular, or partially annular, about the nacelle **50**.

(29) During operation of the turbine engine **10**, a volume of air **58** enters the turbine engine **10**

through an inlet **60** of the nacelle **50** or the fan section **14**. As the volume of air **58** passes across the fan blades **40**, a first portion of air **62**, also referred to as bypass air **62**, is directed or routed into the bypass airflow passage **56**, and a second portion of core air **64**, also referred to as core air **64**, is directed or is routed into the upstream section of the core air flow path **33**, or, more specifically, into the core inlet **20**. The ratio between the first portion of air **62** and the second portion of core air **64** is known as a bypass ratio. In some embodiments, the bypass ratio may be greater than 18:1, enabled by the water system **92**, as detailed further below. The pressure of the core air **64** is increased by the LPC **22**, generating compressed air **65**, and the compressed air **65** is routed through the HPC **24** and further compressed before being directed into the combustor **26**, where the compressed air **65** is mixed with fuel **67** and burned to generate combustion gases **66**, also referred to as combustion gases **66**.

(30) The LPC **22**, the HPC **24**, or both the LPC **22** and the HPC **24** may include one or more stages, with each subsequent stage further compressing the compressed air **65**. The HPC **24** may have a compression ratio greater than 20:1, preferably, in a range of 20:1 to 40:1. The compression ratio is a ratio of a pressure of a last stage of the HPC **24** to a pressure of a first stage of the HPC **24**. The compression ratio greater than 20:1 is enabled by the water system **92**, as detailed further below.

(31) The combustion gases **66** are routed into the HPT **28** and expanded through the HPT **28** where a portion of thermal energy or kinetic energy from the combustion gases **66** is extracted via sequential stages of HPT stator vanes **68** that are coupled to the outer casing **18** and to a plurality of HPT rotor blades **70** that are coupled to the HP shaft **34**, thus, causing the HP shaft **34** to rotate, which supports operation of the HPC **24**. The combustion gases **66** are then routed into the LPT **30** and expanded through the LPT **30**. Here, a second portion of thermal energy or kinetic energy is extracted from the combustion gases **66** via sequential stages of LPT stator vanes **72** that are coupled to the outer casing **18** and to a plurality of LPT rotor blades **74** that are coupled to the LP shaft **36**, thus, causing the LP shaft **36** to rotate, which supports operation of the LPC **22** and rotation of the fan **38** via the gearbox assembly **46**. The HPT **28**, the LPT **30**, or both the HPT **28** and the LPT **30** may have one or more stages. The HPC **24** having a compression ratio in a range of 20:1 to 40:1 results in the HPT **28** having a pressure expansion ratio in a range of 1.5:1 to 4:1 and the LPT **30** having a pressure expansion ratio in a range of 4.5:1 to 28:1.

(32) The combustion gases **66** are subsequently routed through the one or more core exhaust nozzles **32** of the turbo-engine **16** to provide propulsive thrust. Simultaneously with the flow of the core air **64** through the core air flow path **33**, the bypass air **62** is routed through the bypass airflow passage **56** before being exhausted from a fan bypass nozzle **76** of the turbine engine **10**, also providing propulsive thrust. The combustor **26**, the HPT **28**, the LPT **30**, and the one or more core exhaust nozzles **32** at least partially define a hot gas path **78** for routing the combustion gases **66** through the turbo-engine **16**.

(33) As noted above, the compressed air **65** (i.e., the core air **64**) is mixed with the fuel **67** in the combustor **26** to generate a fuel **67** and air mixture, and combusted, generating combustion gases **66**. The fuel **67** can include any type of fuel **67** used for turbine engines, such as, for example, sustainable aviation fuels (SAF) including biofuels, Jet A, or other hydrocarbon fuels. The fuel **67** also may be a hydrogen-based fuel (H.sub.2), and may include blends with hydrocarbon fuels. In some embodiments, the hydrogen fuel may comprise substantially pure hydrogen molecules (i.e., diatomic hydrogen). The fuel **67** may be a cryogenic fuel **67**. For example, in examples with hydrogen fuel **67**, the hydrogen fuel **67** may be stored in a liquid phase at cryogenic temperatures.

(34) The turbine engine **10** includes a fuel system **80** for providing the fuel **67** to the combustor **26**. The fuel system **80** includes a fuel tank **82** for storing the fuel therein, a fuel manifold **90**, and a combustor fuel supply **84**. The fuel tank **82** can be located on an aircraft (not shown) to which the turbine engine **10** is attached. While a single fuel tank **82** is shown in FIG. 1, the fuel system **80** can include any number of fuel tanks **82**. The fuel manifold **90** apportions the flow of fuel from the fuel tank **82** for the combustor **26**. The combustor fuel supply **84** provides fuel **67** from the fuel

manifold **90** to the combustor **26**. The combustor fuel supply **84** includes one or more lines, conduits, pipes, tubes, etc., configured to carry the fuel from the fuel tank **82** to the combustor **26**. The combustor fuel supply **84** also includes a pump **86** to induce the flow of the fuel through the combustor fuel supply **84** to the combustor **26**. In this way, the pump **86** causes the fuel to flow from the fuel tank **82**, through the combustor fuel supply **84**, and into the combustor **26**. Alternatively, the pump **86** may be integrated into the fuel manifold **90**. The fuel system **80** and, more specifically, the fuel tank **82** and the combustor fuel supply **84**, either collectively or individually, may be a fuel source for the combustor **26**.

(35) In some embodiments, for example, when the fuel is a hydrogen fuel, the fuel system **80** includes one or more vaporizers **88** (illustrated as optionally included by dashed lines) and a fuel manifold **90** in fluid communication with the combustor fuel supply **84**. In such an example, the hydrogen fuel is stored in the fuel tank **82** as liquid hydrogen fuel. The one or more vaporizers **88** heat the liquid hydrogen fuel. The one or more vaporizers **88** may be positioned in the flow path of the fuel between the fuel tank **82** and the fuel manifold **90**, and may be located downstream of the pump **86**. Alternatively, the vaporizers **88** may be incorporated into the structure of the fuel manifold **90**. The one or more vaporizers **88** are in thermal communication with at least one heat source, such as, for example, waste heat from the turbine engine **10** and/or from one or more systems of the aircraft (not shown). The one or more vaporizers **88** heat the liquid hydrogen fuel and the liquid hydrogen fuel is converted into a gaseous hydrogen fuel within the one or more vaporizers **88**. The combustor fuel supply **84** directs the gaseous hydrogen fuel into the combustor **26**.

(36) In some embodiments, for example, the fuel manifold **90** is positioned downstream of the one or more vaporizers **88** and the pump **86**. The fuel manifold **90** receives hydrogen fuel in a substantially completely gaseous phase, or in a substantially completely supercritical phase. The fuel manifold **90** provides the flow of fuel to the combustor **26** in a desired manner. More specifically, the fuel manifold **90** provides a desired volume of hydrogen fuel at, for example, a desired flow rate, to one or more fuel injectors that inject the hydrogen fuel into the combustor **26**. The fuel system **80** can include additional components for supplying the fuel from the fuel tank **82** to the combustor **26**, such as, for example, additional pumps, valves, fuel lines, and the like.

(37) The turbine engine **10** includes the water system **92**. The water system **92** provides steam to components of the turbine engine **10** in order to improve or to optimize one or more parameters of engine performance.

(38) The turbine engine **10** depicted in FIG. **1** is by way of example only. In other exemplary embodiments, the turbine engine **10** may have any other suitable configuration. For example, in other exemplary embodiments, the fan **38** may be configured in any other suitable manner (e.g., as a fixed pitch fan) and further may be supported using any other suitable fan frame configuration. Moreover, in other exemplary embodiments, any other suitable number or configuration of compressors, turbines, shafts, or a combination thereof may be provided. In still other exemplary embodiments, aspects of the present disclosure may be incorporated into any other suitable turbine engine, such as, for example, turbofan engines, propfan engines, and/or turboprop engines.

(39) Multiple spools are referenced herein, including the terms fan **38**, low pressure spool, and high pressure spool. The low pressure spool and the high pressure spool may alternatively be referred to as an LP spool **94** and an HP spool **96**, respectively. The term LP spool **94** refers generally to the LPC **22**, the LP shaft **36**, the LPT **30**, and their respective constituent parts. Similarly, the HP spool **96** refers to the HPC **24**, the HP shaft **34**, the HPT **28**, and their respective constituent parts. The fan **38** and its constituent parts may optionally be incorporated into the LP spool **94**. When a gearbox assembly **46** is present, the term LP spool **94** may not include the fan **38** and its constituent parts, as the rotation of the fan **38** is mechanically coupled to the LP spool **94** only at the gearbox assembly **46**. In such cases, the assembly of the fan **38** and its respective constituent parts may be referred to alternatively as a fan spool.

(40) FIG. 2 illustrates schematic view of a turbine engine **200**. The turbine engine **200** is substantially similar to the turbine engine **10** described with respect to FIG. 1, except for the inclusion of multiple thrust pistons that are not shown in FIG. 1, and balance piston cavities that are not shown in FIG. 1. Accordingly, the same reference numerals will be used for components of the turbine engine **200** that are the same as or similar to the components of the turbofan turbine engine **10** as discussed above. The description of these components above also applies to this embodiment, and a detailed description of these components is omitted here. The features and functions of the turbine engine **200** described hereafter, especially as relating to the thrust pistons and balance pistons, are equally applicable to all turbine engines, whether applied to the turbine engine **10** or the turbine engine **200**.

(41) During operation of the turbine engine **200**, several forces act upon various components of the turbine engine **200**, resulting in thrust loads along the longitudinal centerline axis **12**. For example, the core air flow **33**, which becomes compressed air **65** in the LPC **22** and the HPC **24**, subjects each of the LP spool **94** (FIG. 1) and the HP spool **96** (FIG. 1), respectively, to axial loads along the longitudinal centerline axis **12**. Following compression in the LPC **22** and in the HPC **24**, the combustion in the combustor **26** increases pressure of the compressed air **65** and generates combustion gases **66** under pressure higher than the compressed air **65**. The combustion gases **66** flowing first through the HPT **28** and then the LPT **30** apply a thrust load along the longitudinal centerline axis **12**, on each of the HP spool **96** and the LP spool **94**, respectively. The thrust loads due to the operation of the turbine engine **200** act in the forward direction, in the aft direction, or both in the forward direction and the aft direction, depending on multiple factors. One factor may be ambient conditions of the air **58** (e.g., temperature, pressure, humidity, or precipitation). Another factor may be throttle of the turbine engine **200**. More specifically, for the turbine engine **200**, the rate of fuel introduced into the combustor **26** affects the pressure of the combustion gases **66** and consequently impacts axial forces (e.g., thrust loads) on the HP spool **96**, the LP spool **94**, or the HP spool **96** and the LP spool **94**. The forces in each of the LPC **22**, the HPC **24**, the HPT **28**, and the LPT **30** act in different directions at different locations within the respective component. To counteract the various forward acting axial thrust loads and aft acting axial thrust loads, thrust pistons **202** act upon the LP spool **94** and the HP spool **96**. The sum of forces acting on the LP spool **94** due to operation of the LPC **22** and the LPT **30** apply a resultant force on the LP spool **94**. Similarly, the sum of forces acting on the HP spool **96** due to operation of the HPC **24** and the HPT **28** apply a resultant force on the HP spool **96**.

(42) The resultant force on the LP spool **94** and the resultant force on the HP spool **96** depends on the physical state conditions of the core air flow **33**, and the operating conditions of the turbine engine **10**. For example, a higher density of the core air flow **33** may increase the force induced on each of the HPC **24** and the LPC **22**, whereas the rate of fuel consumption may increase force on the HPT **28** and the LPT **30**. Because these forces are variable, the resultant forces on each of the HP spool **96** and the LP spool **94** are also variable.

(43) The turbine engine **200** includes one or more thrust bearings **204** to support thrust loads on the fan **38**, the LP spool **94**, and the HP spool **96**. In addition to the thrust bearings **204**, multiple thrust pistons **202** within the turbine engine **200** support the axial loads of the fan **38**, the axial loads of the LP spool **94** and the axial load of the HP spool **96** on the thrust bearings **204**. The one or more thrust pistons **202** increase the service life of the thrust bearings **204**. Each thrust piston **202** is a cavity including a piston cavity fluid **208** under pressure to apply a compensatory reaction force on the fan **38**, the LP spool **94**, or the HP spool **96** at or near locations where the aforementioned axial forces act on the fan **38**, the LP shaft **36**, or the HP spool **96**, respectively. The piston cavity fluid **208** may be compressed air **65** sourced from the LPC **22**, the HPC **24**, both the LPC **22** and the HPC **24**, or from any other suitable source of pressurized piston cavity fluid **208**. Alternatively, the piston cavity fluid **208** may be water vapor provided by the water system **92** or may be liquid water provided by the water system **92**, as will be discussed later.

(44) The thrust pistons **202** are defined in part by a fixed enclosure portion **210** and in part by a rotating boundary portion **212**. The rotating boundary portion **212** may be a portion of the LP spool **94** or the HP spool **96** that rotates due to operation of the turbine engine **200**, against which the piston cavity fluid **208** acts axially. The fixed enclosure portion **210** and the rotating boundary portion **212** meet at one or more piston cavity interfaces **214**. The piston cavity interfaces **214** may include one or more seals, bearings, or defined gaps to allow for rotation of the rotating boundary portion **212** relative to the fixed enclosure portion **210**. Together, the fixed enclosure portion **210**, the rotating boundary portion **212**, and the piston cavity interfaces **214** generally contain the piston cavity fluid **208** under pressure so as to generate the axial force on the LP spool **94** or the HP spool **96**. The piston cavity interfaces **214** may allow for a defined leak rate of the piston cavity fluid **208** through the piston cavity interfaces **214**, while still maintaining pressure in the thrust piston **202**. Accordingly, the piston cavity interfaces **214** may be a defined gap.

(45) The thrust pistons **202** act similarly upon the LP spool **94** and the HP spool **96** at various locations within the turbine engine **200**. The thrust pistons **202** each may apply a force in the forward direction or in the aft direction on the LP spool **94** or on the HP spool **96**. The direction of the force applied by each thrust piston **202** depends on the location of the thrust piston **202**, in relation to the turbine engine **200** or its internal components, and, thus, which direction the compressed air **65** or the combustion gases **66**, act upon the LP spool **94** or the HP spool **96**.

(46) Other turbine engine inputs may affect the component forces and the total forces on the LP spool **94** and the HP spool **96**. Such inputs include, but are not limited to, direct injection of steam from a water system **92** into the core air flow **33** at one or more locations. Such injection of steam may be performed in order to increase mass flow, thus increasing energy in the HPT **28**, the LPT **30**, or both the HPT **28** and the LPT **30**. Such steam may come from a waste heat recovery system, an onboard water supply, or any other suitable source. Steam injection may be variable, and as such steam injection results in a variable resultant axial force on the LP spool **94**, the HP spool **96**, or both the LP spool **94** and the HP spool **96**.

(47) FIG. 3 shows a detailed schematic view of a forward area **300** of the turbine engine **200**, including two fan (piston) cavities **302**. Each of the fan cavities **302** is defined by a fan rotating boundary portion **304** and by a fan cavity fixed enclosure portions **306**. The fan rotating boundary portions **304** may be one or more components of the fan **38**. Each of four fan cavity interfaces **308** between the fan rotating boundary portion **304** and the fan cavity fixed enclosure portion **306** contains a seal. The LPC **22** provides pressurized a fan piston cavity fluid **310**. The fan piston cavity fluid **310** acts in the aft direction on the fan **38**.

(48) FIG. 4 shows a detailed schematic view of an LPC area **400** of the turbine engine **200**, including a forward booster piston cavity and an aft booster piston cavity, alternatively referred to as an LPC forward cavity **402** and an LPC aft cavity **412**, respectively. The LPC forward cavity **402** is defined by an LPC forward rotating boundary portion **404** and an LPC forward cavity fixed enclosure portion **406**. The LPC forward rotating boundary portion **404** may be a rotating portion of the LPC **22**. Each of two LPC forward cavity interfaces **408** between the LPC forward rotating boundary portion **404**, and the LPC forward cavity fixed enclosure portion **406** contains a seal. The LPC **22** provides a pressurized LPC forward piston cavity fluid **410**. The LPC forward piston cavity fluid **410** acts in the aft direction on the LP spool **94**. The LPC aft cavity **412** is defined by an LPC aft rotating boundary portion **414** and an LPC aft cavity fixed enclosure portion **416**. The LPC aft rotating boundary portion **414** may be a rotating portion of the LPC **22**. Each of two LPC aft cavity interfaces **418** between the LPC aft rotating boundary portion **414** and the LPC aft cavity fixed enclosure portion **416** contains a seal. The LPC **22** provides a pressurized LPC aft piston cavity fluid **420**. The LPC aft piston cavity fluid **420** acts in the forward direction on the LP spool **94**.

(49) FIG. 5 shows a detailed schematic view of a forward area **500** of the turbine engine **200**, including an HPC forward (piston) cavity **502**. The HPC forward cavity **502** is defined by an HPC

forward rotating boundary portion **504** and an HPC forward cavity fixed enclosure portion **506**. The HPC forward rotating boundary portion **504** may be a rotating portion of the HPC **24**. Each of two HPC forward cavity interfaces **508** between the HPC forward rotating boundary portion **504**, and the HPC forward cavity fixed enclosure portion **506** contains a seal. The HPC **24** provides a pressurized HPC forward piston cavity fluid **510**. The HPC forward piston cavity fluid **510** acts in the aft direction on the HP spool **96**.

(50) FIG. **6** shows a detailed schematic view of an intermediate area **600** of the turbine engine **200**, including an HPC aft (piston) cavity **602**, which may alternatively be referred to as a compressor discharge cavity. The HPC aft cavity **602** is defined by an HPC aft rotating boundary portion **604** and an HPC aft cavity fixed enclosure portion **606**. The HPC aft rotating boundary portion **604** may be a rotating portion of the HPC **24**. Each of two HPC aft cavity interfaces **608** between the HPC aft rotating boundary portion **604**, and the HPC aft cavity fixed enclosure portion **606** contains a seal. The HPC **24** provides a pressurized HPC aft piston cavity fluid **610**. The HPC aft piston cavity fluid **610** acts in the forward direction on the HP spool **96**.

(51) The intermediate area **600** shown in FIG. **6** also includes a forward outer seal piston cavity, which may alternatively be referred to as a Forward Outer Seal (FOS) cavity, or an HPT forward (piston) cavity **612**. The HPT forward cavity **612** is defined by an HPT forward rotating boundary portion **614** and an HPT forward cavity fixed enclosure portion **616**. The HPT forward rotating boundary portion **614** may be a rotating portion of the HPT **28**. Each of two HPT forward cavity interfaces **618** between the HPT forward rotating boundary portion **614**, and the HPT forward cavity fixed enclosure portion **616** contains a seal. The HPC **24** provides a pressurized HPT forward piston cavity fluid **620**. The HPT forward piston cavity fluid **620** acts in the aft direction on the HP spool **96**.

(52) The intermediate area **600** shown in FIG. **6** also includes an HPT forward interstage (piston) cavity **622** and an HPT aft interstage (piston) cavity **632**. The forward interstage cavity **622** is defined by a forward interstage rotating boundary portion **624** and a forward interstage cavity fixed enclosure portion **626**. The forward interstage rotating boundary portion **624** may be a rotating portion of the HPT **28**. Each of two forward interstage cavity interfaces **628** between the forward interstage rotating boundary portion **624** and the forward interstage cavity fixed enclosure portion **626** contains a seal. The HPC **24** provides a pressurized forward interstage piston cavity fluid **630**. The forward interstage piston cavity fluid **630** acts in the aft direction on the HP spool **96**. The aft interstage cavity **632** is defined by an aft interstage rotating boundary portion **634** and an aft interstage cavity fixed enclosure portion **636**. The aft interstage rotating boundary portion **634** may be a rotating portion of the HPT **28**. Each of two aft interstage cavity interfaces **638** between the aft interstage rotating boundary portion **634** and the aft interstage cavity fixed enclosure portion **636** contains a seal. The HPC **24** provides a pressurized aft interstage piston cavity fluid **640**. The aft interstage piston cavity fluid **640** acts in the forward direction on the HP spool **96**.

(53) The intermediate area **600** shown in FIG. **6** also includes an HPT aft (piston) cavity **642**. The HPT aft cavity **642** is defined by an HPT aft rotating boundary portion **644** and an HPT aft cavity fixed enclosure portion **646**. The HPT aft rotating boundary portion **644** may be a rotating portion of the HPT **28**. Each of two HPT aft cavity interfaces **648** between the HPT aft rotating boundary portion **644** and the HPT aft cavity fixed enclosure portion **646** contains a seal. The HPC **24** provides a pressurized HPT aft piston cavity fluid **650**. The HPT aft piston cavity fluid **650** acts in the forward direction on the HP spool **96**.

(54) FIG. **7** shows a detailed schematic view of an aft area **700** of the turbine engine, including an LPT forward (piston) cavity **702** and an LPT aft (piston) cavity **712**. The LPT forward cavity **702** is defined by an LPT forward rotating boundary portion **704** and an LPT forward cavity fixed enclosure portion **706**. The LPT forward rotating boundary portion **704** may be a rotating portion of the LPT **30**. Each of two LPT forward cavity interfaces **708** between the LPT forward rotating boundary portion **704** and the LPT forward cavity fixed enclosure portion **706** contains a seal. The

LPC 22 provides a pressurized LPT forward piston cavity fluid 710. The LPT forward piston cavity fluid 710 acts in the aft direction on the LP spool 94. The LPT aft cavity 712 is defined by an LPT aft rotating boundary portion 714 and an LPT aft cavity fixed enclosure portion 716. The LPT aft rotating boundary portion 714 may be a rotating portion of the LPT 30. Each of two LPT aft cavity interfaces 718 between the LPT aft rotating boundary portion 714 and the LPT aft cavity fixed enclosure portion 716 contains a seal. The LPC 22 provides a pressurized LPT aft piston cavity fluid 720. The LPT aft piston cavity fluid 720 acts in the forward direction on the LP spool 94.

(55) FIG. 8 shows a schematic view of an LP balance piston system 800, including an LP balance piston (cavity) 802. As described previously, the thrust pistons 202 (FIG. 2) applied to the fan 38, to the LP spool 94, and to the HP spool 96 each supports a portion of the axial forces applied by the compressed air 65 and the combustion gases 66, flowing through the turbine engine 10 and the turbine engine 200 on the respective fan or spool. The axial forces are variable due to variations in the operation of the turbine engine 200. The thrust pistons 202 alone may not provide appropriate axial reaction force on the LP spool 94 to adequately reduce the resultant axial force on the LP spool 94 to a force that can safely and adequately be supported by the thrust bearings 204. The LP balance piston 802, acting upon the LP spool 94, compensates for the resultant axial thrust load by applying an axial reaction force on the LP spool 94.

(56) To balance the forces on the LP spool 94, one or more of the thrust pistons 202 acting on the LP spool 94, such as the LPT aft cavity 712, may serve as the LP balance piston 802. Alternatively, the LP balance piston 802 may be in the location of the LPC forward cavity 402, the LPC aft cavity 412, or the LPT forward cavity 702. Alternatively, the LP balance piston 802 may be an additional cavity located elsewhere within the turbine engine 200. As described previously with respect to other thrust pistons 202, the LP balance piston cavity includes an LP balance piston rotating boundary portion 804 and an LP balance piston cavity fixed enclosure portion 806 that meet at LP balance piston cavity interfaces 808 to contain a pressurized LP balance piston fluid 810. The LP balance piston cavity interfaces 808 as shown in FIG. 8 are seals, but may alternatively be seals, bearings, gaps, or any combination thereof, through which the LP balance piston fluid 810 passes at a defined acceptable leak rate.

(57) The LP balance piston 802 serves to counteract the resultant axial force on the LP spool 94 through the pressurized LP balance piston fluid 810 that acts axially on the LP balance piston rotating boundary portion 804. The LP balance piston 802 is adjustable, allowing for the pressure of the LP balance piston fluid 810 on the LP spool 94 to be raised or lowered, which adjusts the force applied by the LP balance piston 802 on the LP spool 94 in response to changes in operating conditions of the turbine engine 200. By incorporating the LP balance piston 802, the thrust bearings 204 applied to the LP spool 94 support a lesser axial load applied on the LP spool 94 and minor variation caused by operation of the turbine engine 200, such as vibrations. The thrust bearings 204 applied to the LP spool 94 are better able to support such lesser axial loading, and, as such, have increased performance, increased service life, or both increased performance and increased service life relative to similar thrust bearings 204, as would be applied in an otherwise equivalent engine without the LP balance piston 802.

(58) Additionally, the LP balance piston rotating boundary portion 804 may include one or more passages 812 allowing for liquid or gaseous LP balance piston fluid 810 to pass through the LP balance piston rotating boundary portion 804, into a secondary cavity 814, exposing the LP balance piston fluid 810 to additional components of the turbine engine 200. The secondary cavity 814 may be one or more secondary cavities 814. Such exposed components may include the LP shaft 36, one or more LPT rotor disks 816, or other components, depending on the location and orientation of the one or more passages 812.

(59) The LP balance piston system 800 includes multiple components for providing water to the LP balance piston 802. A water supply 819 provides a source of liquid water 820 to the LP balance piston 802. A water line 822 provides the water 820 from the water supply 819 to at least one water

pump **824**. The water pump **824** provides a flow of the water **820** through the water line **822** to the LP balance piston **802**. The water line **822** includes one or more check valves **826** disposed between the water pump **824** and the LP balance piston **802** that prevent a reverse flow of the water **820** (liquid or vapor) through the water line **822**. The water line **822** further connects to the LP balance piston **802**, providing the piston with a source of liquid water **820**.

(60) Once introduced to the LP balance piston **802**, the water **820** becomes the LP balance piston fluid **810** and absorbs heat from adjacent components. Specifically, the LP balance piston fluid **810** absorbs heat from the relatively hotter components, including the LP balance piston rotating boundary portion **804**, the LP shaft **36**, the LP balance piston cavity fixed enclosure portion **806**, and the LP balance piston cavity interfaces **808**, whereupon the LP balance piston fluid **810** evaporates, converting partially or entirely from liquid water to steam, by absorbing heat from the adjacent components. Therefore the LP balance piston fluid **810** may be supercritical steam or may be subcritical steam. Subcritical steam is also referred to as two-phase steam.

(61) The water system **92** may include a waste heat recovery system, a dedicated onboard water supply, or any other suitable water source. By controlling the flow rate of the water **820** into the LP balance piston **802**, the axial pressure, and therefore the axial force applied on the LP balance piston rotating boundary portion **804** is adjustable. The pressure applied on the LP balance piston rotating boundary portion **804** is additionally affected by the rate of temperature difference between adjacent components, such as, for example, the LP balance piston rotating boundary portion **804** to the LP balance piston fluid **810**, as well as by the flow rate of the LP balance piston fluid **810** through the LP balance piston **802**.

(62) As discussed, the LP balance piston **802** may also contain the one or more passages **812** that allow flow of the LP balance piston fluid **810** through the LP balance piston rotating boundary portion **804**, and into the one or more secondary cavities **814**. Upon entering the secondary cavity **814**, the LP balance piston fluid **810** is further exposed to heated elements of the turbine engine **200**, including the LPT rotor disks **816**. Upon exposure to the LPT rotor disks **816**, the LP balance piston fluid **810** absorbs additional heat, now from the LPT rotor disks **816**. In the case of two-phase steam as the LP balance piston fluid **810**, the LP balance piston fluid **810** will increase in quality, potentially becoming supercritical, and, in the case of supercritical steam as the LP balance piston fluid **810**, the LP balance piston fluid **810** will increase in temperature. Heat transfers from the components within the secondary cavity **814** to the LP balance piston fluid **810**. This heat transfer occurs whether the LP balance piston fluid **810** is two-phase steam or supercritical steam and lowers the temperature of the components of the secondary cavity **814**.

(63) In order to facilitate heat transfer, the LP rotor disks **816** may be attached to the LP balance piston rotating boundary portion **804**, as shown in FIG. **8**. By being attached to the LP balance piston rotating boundary portion **804**, the secondary cavity **814** may be separated into the one or more secondary cavities **814**. The one or more secondary cavities **814** may fluidly communicate with one another via passages (not shown) through the LP rotor disks **816**.

(64) Exiting the secondary cavity **814**, the LP balance piston fluid **810** further flows through one or more passages **818** in the LP spool **94** to the LPT **30**, wherein the LP balance piston fluid **810** mixes with the combustion gases **66**. Such introduction of steam into the LPT **30** may increase the overall energy generation of the LPT **30** for the reasons previously discussed. Multiple passages **818** may be required to allow the LP balance piston fluid **810** to flow out of the each of the one or more secondary cavities **814**, when there is more than one secondary cavity.

(65) The conversion of the LP balance piston fluid **810** from liquid to vapor creates pressure in the LP balance piston **802** to counteract the resultant force on the LP spool **94** from the turbine engine **200** operation, including the thrust pistons **202**. Upon contact with heated surfaces of, for example, the LP balance piston rotating boundary portion **804**, heat transfers from the LP balance piston rotating boundary portion **804** to the LP balance piston fluid **810**, causing the LP balance piston fluid **810** to evaporate to steam, expanding the LP balance piston fluid and increase in pressure. The

pressure applied by the expanded LP balance piston fluid **810** acts upon the LP balance piston rotating boundary portion **804** and supports a portion of the axial load on the LP spool **94**.

(66) As described above with respect to the thrust pistons **202**, the LP balance piston cavity interfaces **808** may be gaps. The LP balance piston cavity interfaces **808** allow a defined leak rate of the LP balance piston fluid **810** through the LP balance piston **802** while still maintaining adequate pressure in the LP balance piston **802** to axially support the LP spool **94**. The LP balance piston fluid **810** provided by the water supply **819** is cooler than the components of the LP balance piston **802** (e.g., the LP balance piston rotating boundary portion **804**). By flowing continuously through the LP balance piston **802** and out through the LP balance piston cavity interfaces **808**, the LP balance piston fluid **810** provides continuous heat transfer away from the LP balance piston **802**. The LP balance piston fluid **810** cools the LP spool **94** by conductive heat transfer. Heat in the LP spool **94** travels from the warmer portions, including the LPT **30**, to the cooled portions including the LP balance piston rotating boundary portion **804**.

(67) To maintain continuous heat transfer to the LP balance piston fluid **810**, a continuous flow of the LP balance piston fluid **810** is necessary. To facilitate such flow and such continuous heat transfer, the LP balance piston rotating boundary portion **804** includes one or more passages **812**, allowing for the LP balance piston fluid **810** to pass out of the LP balance piston **802** and into the secondary cavity **814**, wherein the LP balance piston fluid **810** may contact and further absorb heat from additional portions of the LP spool **94**. Such gaps may be present in addition to, or as an alternative to, a defined leak through the LP balance piston cavity interfaces **808**. Such additional portions of the LP spool **94** include the LPT rotor disks **816** and the LP shaft **36**. After flowing through the secondary cavity **814** and having further absorbed heat from the additional portions of the LP spool **94**, the LP balance piston fluid **810** flows out of the secondary cavity **814** through the one or more passages **818**.

(68) The passages **818** allow for the outflow of the LP balance piston fluid **810** from the secondary cavity **814** to the path of the combustion gases **66** in the LPT **30**. Steam introduced into the LPT **30** will increase mass flow through the LPT **30**, thus increasing the energy generated therein. Additionally, further heat transfer to the gaseous LP balance piston fluid **810** from the LPT **30** will lower the temperature of the combustion gases **66** within the LPT **30**, which reduces the risk of damage to the LPT components, such as LPT rotor blades **828** and LPT stator vanes **829**.

(69) Alternatively, the LP balance piston fluid **810** may flow through passages from the secondary cavity **814**, to be expelled from the turbine engine **200**, bypassing the core air flow **33** (not shown). Alternatively, in the case of a closed loop water system **92**, the LP balance piston fluid **810** may recondense in a condenser (not shown) and return to the water supply **819** as liquid water **820**. All of the passages **812** and the passages **818** are sized to ensure a greater pressure within the LP balance piston **802** than within the secondary cavity **814**, allowing the LP balance piston **802** to continue to apply axial force to the LP spool **94**.

(70) The at least one water pump **824** causes liquid water from the water supply **819** to flow through the water line **822**, whereupon reaching the LP balance piston **802**, the liquid water becomes the LP balance piston fluid **810**. A benefit of pumping liquid water **820** into the LP balance piston **802** and converting to a pressurized gas once in the LP balance piston **802** is that, due to its nearly incompressible state, liquid water may be pumped at lower pressures than the equivalent gas (water vapor or air) pump would require. An additional benefit of causing liquid or gaseous water to flow into the LP balance piston **802** is that, by absorbing thermal energy in the phase change from liquid to vapor within the LP balance piston **802**, the LP balance piston fluid **810** cools adjacent components such as the LP balance piston rotating boundary portion **804**. Cooling such adjacent components may contribute to prolonged life of the turbine engine **200** or improved operational performance of the turbine engine **200**.

(71) The water line **822** may include the one or more check valves **826**, also known as non-return valves, and one or more flow control valves **827**. As discussed, the check valves **826** ensure that the

heated or pressurized water vapor does not flow backwards, into the water pump **824**, which could damage the water pump **824**. Additionally, the one or more check valves **826** ensure that a backflow of liquid water or steam cannot introduce heat to portions of the turbine engine **200** or water system **92** that are unable to tolerate such heat. Further, the one or more check valves **826** ensure the proper direction of flow of the LP balance piston fluid **810** in the LP balance piston **802**, which helps ensure the heat transfer to the LP balance piston fluid **810**.

(72) The LP balance piston system **800** includes a controller **830**. As the resultant axial force on the LP spool **94** is variable, as previously discussed, the controller **830** controls the speed of the water pump **824** and adjusts the flow control valve **827** in order to control the outlet flow of the water **820** from the water pump **824** to the LP balance piston **802**. The controller controls the flow of the water **820** in such a manner as to adjust the pressure applied by the LP balance piston fluid **810** on the LP balance piston rotating boundary portion **804** and controlling the axial force of the LP balance piston **802** on the LP spool **94**. Such control of the axial force applied by the LP balance piston **802** on the LP spool **94** is necessary to adjust to changing operational conditions of the turbine engine **200**. In addition to sending control data to the water pump **824** and to the flow control valve **827**, the controller **830** receives data from the water pump **824** and from the flow control valve **827**. The data from the water pump **824** and the data from the flow control valve **827** may include pressure and/or flow rates therethrough. The controller **830** may detect certain system faults, such as, for example, a broken water line **822** resulting in a loss of water **820**, by monitoring the data from the water pump **824** and/or the flow control valve **827**.

(73) The controller **830** may be an engine controller. The controller **830** is configured to operate various aspects of the turbine engine **200**, including, in the embodiments discussed herein, the LP balance piston system **800** and its constituent components. The controller **830** may be a Full Authority Digital Engine Control (FADEC). In this embodiment, the controller **830** is a computing device having one or more processors **832** and one or more memories **834**. The processor **832** may be any suitable processing device, including, but not limited to, a microprocessor, a microcontroller, an integrated circuit, a logic device, a programmable logic controller (PLC), an application-specific integrated circuit (ASIC), and/or a Field Programmable Gate Array (FPGA). The memory **834** may include one or more computer-readable media, including, but not limited to, non-transitory computer-readable media, a computer-readable non-volatile medium (e.g., a flash memory), a RAM, a ROM, hard drives, flash drives, and/or other memory devices.

(74) The memory **834** may store information accessible by the processor **832**, including computer-readable instructions that may be executed by the processor **832**. The instructions may be any set of instructions or a sequence of instructions that, when executed by the processor **832**, causes the processor **832** and the controller **830** to perform operations. In some embodiments, the instructions may be executed by the processor **832** to cause the processor **832** to complete any of the operations and functions for which the controller **830** is configured, as described herein. The instructions may be software written in any suitable programming language or may be implemented in hardware. Additionally or alternatively, the instructions may be executed in logically and/or virtually separate threads on the processor **832**. The memory **834** may further store data that may be accessed by the processor **832**. The memory **834** may apply certain rules or control logic, which control the apportionment of fuel and steam. Such rules may apply to engine power, emissions targets, engine safety parameters, engine efficiency, or any other such relevant measurable output.

(75) The technology discussed herein makes reference to computer-based systems and actions taken by, and information sent to and from, computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between components and among components. For instance, processes discussed herein may be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications may be implemented on a single system or

distributed across multiple systems. Distributed components may operate sequentially or in parallel.

(76) Increasing the water flow rate in the water system **92** may increase the pressure of the LP balance piston fluid **810** in the LP balance piston **802**, applying greater axial pressure (therefore, a greater axial force) on the LP spool **94**. The water pump **824**, and the one or more flow control valves **827** may be adjusted to allow a greater or a lesser flow rate of water through the LP balance piston **802**. A greater flow rate may allow for more heat transfer from the turbine engine **200** components, by virtue of maintaining a greater temperature difference between the LP balance piston fluid **810** and adjacent components, such the LP balance piston rotating boundary portion **804**. A lesser flow rate may allow for a greater pressure in the LP balance piston fluid **810**, therefore, a greater pressure on the LP balance piston rotating boundary portion **804**, and therefore a greater axial force applied on the LP spool **94** by the LP balance piston **802**.

(77) While FIG. **8** and the accompanying description reference the LP balance piston system **800** including the LP balance piston **802** that acts upon the LP spool **94**, the foregoing description with respect to the LP balance piston system **800** may additionally or alternatively apply to a balance piston acting on the HP spool **96**, with the same water system **92** or an additional water system that similarly serves to balance the axial forces on the HP spool **96** resulting from the internal forces of the compressed air **65**, the combustion gases **66**, and the external forces applied on the HP spool **96** by the thrust pistons **202**.

(78) While the LP balance piston **802** as shown in FIG. **8** acts upon the LP spool **94** in the forward direction, compensating for an aft thrust load on the shaft, an LP balance piston or an HP balance piston may be applied in such a manner as to act in the aft direction, to compensate for a forward load on the LP spool **94** or HP spool **96**. Additionally, an LP spool **94** or an HP spool **96** may have both forward-acting pistons and aft-acting balance pistons, if the operational design of the engine is such that resultant axial forces to be compensated by the respective balance piston, are expected to reverse.

(79) The foregoing description with respect to the LP balance piston system **800** of FIG. **8** may alternatively be applied to any of the thrust pistons **202** as described with respect to FIG. **2**. In such cases, the water system **92** provides water that changes phase in the thrust piston **202** in the manner as described above with respect to the LP balance piston **802**, providing a pressure in the thrust piston **202**, and in turn a force on the LP spool **94** or the HP spool **96**. In such cases, the thrust pistons **202** rather than applying a substantially static load, may apply a variable load by controlling pump speed and flow control valves as described above with respect to the LP balance piston system **800**.

(80) FIG. **9** shows a schematic view of a balance piston system **900** including both a first balance piston and a second balance piston. The first balance piston may be an LP balance piston **922** and the second balance piston may be an HP balance piston **924**. The balance system includes a water supply **902**, to provide water **906** through one or more water lines **904** to one or more water pumps **908**. The one or more water pumps **908** generates a pressure, causing the water **906** to further flow through the water line **904** to the LP balance piston **922** and the HP balance piston **924**. A check valve **910** prevents backflow in the water line **904** and a flow control valve **912** controls the rate of flow of the water **906** to the LP balance piston **922** and the HP balance piston **924**. Such a system allows for a single source of the water **906** to supply both the LP balance piston **922** and the HP balance piston **924**. A water supply line **914** with an LP flow control valve **918** located therein, supplies the water **906** to the LP balance piston **922**, and may be independently adjusted by control of the LP flow control valve **918**. Similarly, a water supply line **914** with an HP flow control valve **920** located therein, supplies the water **906** to the HP balance piston **924**, and may be independently adjusted by control of the HP flow control valve **920**.

(81) The balance piston system **900** also includes a controller **930**. The controller **930** contains a processor **932** and a memory **934**. The foregoing description of the controller **830**, the processor

832, and the memory **834** applies similarly to the controller **930**, the processor **932**, and the memory **934** respectively, for controlling a balance piston system **900**. The controller **930** controls the pressure in the LP balance piston **922** and the pressure in the HP balance piston **924**, by controlling the speed of the water pump **908**, and by adjusting the settings of the flow control valve **912**, the LP flow control valve **918**, and the HP flow control valve **920**. This control controls the rate and the pressure of the flow of the water **906** in the water line **904** to each of the LP balance piston **922** and the HP balance piston **924**.

(82) Further, the controller receives data from the water pump **908**, the flow control valve **912**, the LP flow control valve **918**, and the HP flow control valve **920**, which may be used for the detection of certain system faults, such as, for example, a broken water line **904** or a broken water supply line **914**, resulting in a loss of the water **906**, by monitoring the data from the water pump **908**, the flow control valve **912**, the LP flow control valve **918**, and the HP flow control valve.

(83) FIG. **10** shows a schematic view of an alternative balance piston system **1000** including the LP balance piston **922** and the HP balance piston **924**. The balance piston system **1000** is substantially similar to the balance piston system **900**, described with respect to FIG. **9**, with the addition of an evaporator **1006** (a heat exchanger) that generates steam **1002** that flows to the LP balance piston **922**, to the HP balance piston **924**, and to a steam turbine **1010**. The steam turbine **1010** further provides steam to the combustor **26**, to one or more of the thrust pistons **202**, or both to the combustor **26** and to one or more of the thrust pistons **202**. Accordingly, the same reference numerals will be used for components of the balance piston system **1000** that are the same as or similar to the components of the balance piston system **900** discussed above. The description of these components above also applies to this embodiment, and a detailed description is omitted here.

(84) Liquid water **906** is provided by the water supply **902** through the water line **904** to one or more water pumps **908**. The water pump **908** pressurizes the liquid water **906**, inducing a further flow of the liquid water **906** through the water line to the evaporator **1006**. The evaporator **1006** converts the liquid water **906** to the steam **1002**. The evaporator **1006** operates by receiving a flow of the combustion gases **66** from the LPT **30** or the HPT **28**, at a higher temperature than the water **906**. Heat transfer from the core air flow **33** to the liquid water **906** occurs within the evaporator (heat exchanger) **1006**. The heat transfer, and resulting increase of temperature of the liquid water **906**, causes conversion to the steam **1002**. Following the evaporator **1006**, the steam **1002** flows in a steam line **1004** first to a first steam control valve, or a first steam manifold **1008**. The first steam manifold **1008** directs a portion of the steam **1002** to another steam line **1004** leading to the steam turbine **1010** and directs the remainder of the steam **1002** to other steam lines **1004** leading to the LP balance piston **922** and to the HP balance piston **924**.

(85) The steam lines **1004** provide the steam **1002** to the LP balance piston **922** and the HP balance piston **924**. An LP flow control valve **1014** and an HP flow control valve **1016** independently control the flow rate of the steam **1002** in the steam lines **1004** to each of the LP balance piston **922** and the HP balance piston **924**, respectively. This control allows for independent control of the force applied by each of the LP balance piston **922** and the HP balance piston **924**.

(86) The balance piston system **1000** also includes a controller **1030**. The controller **1030** contains a processor **1032** and a memory **1034**. The foregoing description of the controller **830**, the processor **832**, and the memory **834** applies similarly to the controller **1030**, the processor **1032**, and the memory **1034**, respectively, for controlling the balance piston system **1000**. The controller **1030** controls the pressure in the LP balance piston **922** and the pressure in the HP balance piston **924**, by controlling the speed of the water pump **908**, and by adjusting the settings of the flow control valve **912**, the LP flow control valve **1014**, and the HP flow control valve **1016**. The controller **1030** also controls the settings of the first steam manifold **1008**, which adjusts the apportionment of the steam **1002** to the steam turbine **1010**. The controller **1030** also controls a second control valve, or second steam manifold **1012**, to adjust the apportionment of the steam **1002** output by the steam turbine **1010** to the combustor **26**, to the one or more of the thrust pistons

202, or to both the combustor **26** and to the one or more of the thrust pistons **202**. The controller **1030** controls the rate and the pressure of the flow of the water **906** in the water line **904** to the evaporator **1006** and the flow of the steam **1002** in the steam lines **1004** to each of the LP balance piston **922** and the HP balance piston **924**.

(87) The controller **1030** may additionally detect system failures, such as, for example, a broken water line **904** resulting in a loss of the water **906**, through data received from the water pump **908**, and/or the flow control valve **912**. Additionally, the controller may detect system failures, such as, for example a broken steam line **1004** resulting in a loss of the steam **1002**, through data received from the first steam manifold **1008**, the LP flow control valve **1014**, the HP flow control valve **1016**, and/or the steam control valve **1012**.

(88) The steam turbine **1010** extracts thermal energy, kinetic energy, or both thermal energy and kinetic energy from the steam **1002**. The energy may contribute to rotation of the LP spool **94** or the HP spool, or may be converted into an alternative form of energy (e.g., electrical energy) for use outside of the turbine engine. Upon exiting the steam turbine **1010**, the steam **1002** flows through the second steam manifold **1012**. The second steam manifold apportions the steam **1002** flowing from the steam turbine **1010**, through additional separate steam lines **1004** to the combustor **26**, to one or more of the thrust pistons **202**, or to both the combustor **26** and to one or more of the thrust pistons **202**. The introduction of the steam **1002** to the combustor **26** may enhance performance of the HPT **28**, the LPT **30**, or both the HPT **28** and the LPT **30**. The provision of the steam to one or more of the thrust pistons **202** may provide an alternative source of the piston cavity fluid **208** for support of the LP spool **94** or the HP spool **96**.

(89) While the embodiments shown include a single balance piston on each of the LP spool **94** and the HP spool **96**, embodiments are contemplated with more than one balance piston on the LP spool **94**, the HP spool **96**, or both the LP spool **94** and the HP spool **96**. In such embodiments, one or more of the thrust pistons **202** may function as balance piston cavities, with controlled water flow that in turn controls the pressure applied by the piston cavity fluid on the respective rotating boundary portion.

(90) Further aspects are provided by the subject matter of the following clauses.

(91) A turbine engine comprises a spool, the spool comprising a compressor that compresses core air, a turbine, and a shaft connecting the compressor and the turbine such that the compressor and the turbine rotate together, a fuel supply, a combustor located downstream of the compressor that receives the core air and the fuel supply and combusts the core air and the fuel supply to generate combustion gases, the combustion gases flowing to the turbine, causing rotation of the turbine, the core air, the combustion gases, the compressor, and the turbine generating a first axial force on the spool, a balance piston that applies a second axial force on the spool to balance at least a portion of the first axial force, the balance piston comprising a rotating boundary portion comprising a rotating portion of the spool, and a fixed enclosure portion that does not rotate, the rotating boundary portion and the fixed enclosure portion defining a balance piston cavity, and a water system providing water to the balance piston cavity the water in the balance piston cavity is a balance piston fluid, and the balance piston fluid absorbs heat from the balance piston, causing an expansion and a pressure increase in the balance piston fluid generating the second axial force.

(92) The turbine engine of the preceding clause, the water system comprising a water supply supplying the water to the water system, water supply lines connecting the water supply to the balance piston, a water pump causing a flow of the water through the water supply lines, a flow control valve controlling a rate of the flow of the water to the balance piston, and a controller controlling operation of the water pump and the flow control valve to control the second axial force by controlling the rate of the flow of the water to the balance piston.

(93) The turbine engine of any preceding clause, wherein the balance piston fluid is two-phase steam.

(94) The turbine engine of any preceding clause, wherein the balance piston fluid is supercritical

steam.

(95) The turbine engine of any preceding clause, the balance piston further comprising at least one interface between the rotating boundary portion and the fixed enclosure portion, the at least one interface allowing for a defined leak rate of the balance piston fluid from the balance piston cavity, to allow for continuous flow of water through the balance piston cavity.

(96) The turbine engine of any preceding clause, wherein the spool is a low pressure spool, the compressor is a low pressure compressor, the turbine is a low pressure turbine, the shaft is a low pressure shaft, the first axial force is a low pressure first axial force, the balance piston is a low pressure balance piston, the rotating boundary portion is a low pressure rotating boundary portion, the fixed enclosure portion is a low pressure fixed enclosure portion, the balance piston cavity is a low pressure balance piston cavity, the balance piston fluid is a low pressure balance piston fluid, the second axial force is a low pressure second axial force, the turbine engine further comprises a high pressure spool, the high pressure spool comprises a high pressure compressor disposed between the low pressure compressor and the combustor that further compresses the core air from the low pressure compressor before the core air flows into the combustor, a high pressure turbine disposed between the combustor and the low pressure turbine, caused to rotate by the combustion gases prior to the combustion gases flowing into the low pressure turbine, and a high pressure shaft connecting the high pressure compressor and the high pressure turbine such that the high pressure compressor and the high pressure turbine rotate together, the core air, the high pressure compressor, and the high pressure turbine generating a high pressure first axial force on the spool, a high pressure balance piston that applies a high pressure second axial force on the high pressure spool to balance at least a portion of the high pressure first axial force, the high pressure balance piston comprises a high pressure rotating boundary portion that comprises a rotating portion of the high pressure spool, and a high pressure fixed enclosure portion that does not rotate, the high pressure rotating boundary portion and the high pressure fixed enclosure portion defining a high pressure balance piston cavity, wherein the water system provides water to the high pressure balance piston cavity, the water in the high pressure balance piston cavity is a high pressure balance piston fluid, the high pressure balance piston fluid absorbs heat from the high pressure balance piston, causing an expansion and a pressure increase in the high pressure balance piston fluid, generating the high pressure second axial force.

(97) The turbine engine of the preceding clause, the water system further comprising a water supply supplying the water to the water system, water supply lines connecting the water supply to the high pressure balance piston and the low pressure balance piston, a water pump causing a flow of the water through the water supply lines, a low pressure flow control valve controlling a rate of the flow of the water to the low pressure balance piston, a high pressure flow control valve controlling a rate of the flow of the water to the high pressure balance piston, and a controller controlling operation of the water pump, the low pressure flow control valve and the high pressure flow control valve in order to control the rate of the flow of the water to the low pressure balance piston and to the high pressure balance piston, to control the low pressure second axial force and the high pressure second axial force, respectively

(98) The turbine engine of any preceding clause, wherein the water system further comprises a liquid water supply and an evaporator that converts the liquid water supply into a steam supply that is introduced into the balance piston.

(99) The turbine engine of the preceding clause, the water system further comprising a steam turbine, wherein the steam supply flows to the steam turbine, the steam turbine extracts a portion of thermal energy, kinetic energy, or thermal energy and kinetic energy from the steam supply.

(100) The turbine engine of the preceding clause, wherein the steam supply further flows to the combustor.

(101) The turbine engine of any preceding clause, wherein the steam supply further flows to a piston cavity.

- (102) The turbine engine of any preceding clause, the shaft defining a secondary cavity, and the rotating boundary portion further comprises at least one passage therethrough, wherein the balance piston fluid flows through the passage and into the secondary cavity.
- (103) The turbine engine of the preceding clause, wherein the passage is a first passage, the shaft further comprises at least one second passage therethrough, and the balance piston fluid flows from the secondary cavity through the second passage into the turbine.
- (104) The turbine engine of any preceding clause, wherein the secondary cavity comprises turbine rotor disks that transfer heat to the balance piston fluid in the secondary cavity, to cool the turbine rotor disks.
- (105) The turbine engine of any preceding clause, wherein the secondary cavity is a plurality of secondary cavities, the passage is a plurality of passages, and at least one of the plurality of passages connects to each of the plurality of secondary cavities allowing the balance piston fluid to flow through into each of the plurality of secondary cavities.
- (106) The turbine engine of the preceding clause, wherein the plurality of passages is a plurality of first passages, the shaft further comprises a plurality of second passages therethrough, at least one of the plurality of second passages connecting to each of the plurality of secondary cavities, and the balance piston fluid flows from the secondary cavities through the respective second passages into the turbine.
- (107) A method of supporting axial force on the turbine engine of any preceding clause, the method comprising providing water to the balance piston, the water becoming the balance piston fluid, transferring heat from the balance piston to the balance piston fluid to expand the balance piston fluid and to increase pressure of the balance piston fluid, and generating the second axial force.
- (108) The method of the preceding clause, wherein the water system further comprises a water supply supplying the water to the water system, water supply lines connecting the water supply to the balance piston, a water pump causing a flow of the water through the water supply lines, a flow control valve controlling a rate of the flow of the water to the balance piston, and a controller, and the method further comprises controlling the water pump and the flow control valve in order to control the rate of the flow of the water to the balance piston, to control the second axial force.
- (109) The method of any preceding clause, wherein the water is in a liquid phase until the water enters the balance piston cavity, becoming two-phase steam through heat transfer from the balance piston cavity.
- (110) The method of any preceding clause, wherein the water is in a liquid phase until the water enters the balance piston cavity becoming supercritical steam through heat transfer from the balance piston cavity.
- (111) The method of any preceding clause, the balance piston further comprising at least one interface between the rotating boundary portion and the fixed enclosure portion, and the method further comprising leaking the balance piston fluid from the balance piston cavity, through the at least one interface, at a defined rate.
- (112) The method of any preceding clause, wherein the spool is a low pressure spool, the compressor is a low pressure compressor, the turbine is a low pressure turbine, the shaft is a low pressure shaft, the balance piston is a low pressure balance piston, the rotating boundary portion is a low pressure rotating boundary portion, the fixed enclosure portion is a low pressure fixed enclosure portion, the balance piston fluid is a low pressure balance piston fluid, the second axial force is a low pressure second axial force, the turbine engine further comprises a high pressure spool, the high pressure spool comprises a high pressure compressor disposed between the low pressure compressor and the combustor that further compresses the core air from the low pressure compressor before the core air flows into the combustor, a high pressure turbine disposed between the combustor and the low pressure turbine, the high pressure turbine caused to rotate by the combustion gases prior to flowing into the low pressure turbine, and a high pressure shaft connecting the high pressure compressor and the high pressure turbine such that the high pressure

compressor and the high pressure turbine rotate together, the core air, the high pressure compressor, and the high pressure turbine generating a high pressure first axial force on the spool, a high pressure balance piston that applies a high pressure second axial force on the high pressure spool to balance at least a portion of the high pressure first axial force, the high pressure balance piston comprises a high pressure rotating boundary portion comprises a rotating portion of the high pressure spool, and a high pressure fixed enclosure portion that does not rotate, the high pressure rotating boundary portion and the high pressure fixed enclosure portion defining a high pressure balance piston cavity, and the method further comprises providing water to the high pressure balance piston cavity, such that the water in the high pressure balance piston cavity, the water becomes a high pressure balance piston fluid, transferring heat from the high pressure balance piston to the high pressure balance piston fluid, expanding the high pressure balance piston fluid and increasing pressure of the high pressure balance piston fluid, and generating the high pressure second axial force.

(113) The method of the preceding clause, wherein the water system further comprises a water supply supplying the water to the water system, water supply lines connecting the water supply to the low pressure balance piston and to the high pressure balance piston, a water pump causing a flow of the water through the water supply lines, a low pressure flow control valve controlling a rate of the flow of the water to the low pressure balance piston, a high pressure flow control valve controlling a rate of the flow of the water to the high pressure balance piston, and a controller, and the method further comprises controlling the water pump, the low pressure flow control valve and the high pressure flow control valve in order to control the rate of the flow of the water to the low pressure balance piston and to the high pressure balance piston, to control the low pressure second axial force and the high pressure second axial force, respectively.

(114) The method of any preceding clause, wherein the water system further comprises an evaporator, and the water is a liquid water supply prior to entering the evaporator, and the method further comprises converting the liquid water supply into a steam supply in the evaporator and introducing the steam supply into the balance piston.

(115) The method of the preceding clause, wherein the water system further comprises a steam turbine, and the method further comprises causing the steam supply to flow to the steam turbine, extracting a portion of thermal energy, kinetic energy, or thermal energy and kinetic energy from the steam supply.

(116) The method of the preceding clause, further comprising causing the steam supply to flow to the combustor after flowing through the steam turbine.

(117) The method of any preceding clause, further comprising causes the steam supply to flow to a piston cavity after flowing through the steam turbine.

(118) The method of any preceding clause, the shaft further comprises a secondary cavity, and the rotating boundary portion further comprises at least one passage therethrough, the method further comprises flowing the balance piston fluid through the passage and into the secondary cavity and transferring heat from the secondary cavity to the balance piston fluid.

(119) The method of the preceding clause, wherein the secondary cavity comprises turbine rotor disks, the method further comprises transferring heat from the turbine rotor disks to the balance piston fluid in the secondary cavity, to cool the turbine rotor disks.

(120) The method of the preceding clause, wherein the passage is a first passage, and the shaft further comprises at least one second passage therethrough, and the method further comprises flowing the balance piston fluid from the secondary cavity through the passage into the turbine.

(121) The method of any preceding clause, wherein the secondary cavity is a plurality of secondary cavities and the passage is a plurality of passages, at least one of the plurality of passages connecting to each of the secondary cavities, the method further comprises causing the balance piston fluid to flow to the plurality of secondary cavities and transferring heat from each of the plurality of secondary cavities to the balance piston fluid in the respective secondary cavity.

(122) The method of the preceding clause, wherein the plurality of passages is a plurality of first passages, the shaft further comprises a plurality of second passages therethrough, at least one of the plurality of second passages connecting to each of the plurality of secondary cavities, and the method further comprises causing the balance piston fluid to flow from each of the plurality of secondary cavities through the corresponding at least one of the plurality of secondary passages into the turbine.

(123) Although the foregoing description is directed to the preferred embodiments, it is noted that other variations and modifications will be apparent to those skilled in the art, and may be made without departing from the disclosure. Moreover, features described in connection with one embodiment may be used in conjunction with other embodiments, even if not explicitly stated above.

Claims

1. A turbine engine comprising: a spool comprising: a compressor that compresses core air; a turbine; and a shaft connecting the compressor and the turbine such that the compressor and the turbine rotate together; a fuel supply; a combustor located downstream of the compressor, the combustor receiving the core air and the fuel supply, and combusts the core air and the fuel supply to generate combustion gases, the combustion gases flowing to the turbine, causing rotation of the turbine, the core air, the combustion gases, the compressor, and the turbine generating a first axial force on the spool; a balance piston that applies a second axial force on the spool to balance at least a portion of the first axial force, the balance piston comprising: a rotating boundary portion comprising a rotating portion of the spool; and a fixed enclosure portion that does not rotate, the rotating boundary portion and the fixed enclosure portion defining a balance piston cavity; and a water system providing water to the balance piston cavity, wherein the water in the balance piston cavity is a balance piston fluid, the balance piston fluid absorbing heat from the balance piston, causing an expansion and a pressure increase in the balance piston fluid generating the second axial force.

2. The turbine engine of claim 1, wherein the balance piston fluid is two-phase steam or supercritical steam.

3. The turbine engine of claim 1, wherein the spool is a low pressure spool, the compressor is a low pressure compressor, the turbine is a low pressure turbine, the shaft is a low pressure shaft, the first axial force is a low pressure first axial force, the balance piston is a low pressure balance piston, the rotating boundary portion is a low pressure rotating boundary portion, the fixed enclosure portion is a low pressure fixed enclosure portion, the balance piston cavity is a low pressure balance piston cavity, the balance piston fluid is a low pressure balance piston fluid, the second axial force is a low pressure second axial force, and the turbine engine further comprises: a high pressure spool comprising: a high pressure compressor disposed between the low pressure compressor and the combustor that further compresses the core air from the low pressure compressor before the core air flows into the combustor; a high pressure turbine disposed between the combustor and the low pressure turbine, the high pressure turbine caused to rotate by the combustion gases prior to the combustion gases flowing into the low pressure turbine; and a high pressure shaft connecting the high pressure compressor and the high pressure turbine such that the high pressure compressor and the high pressure turbine rotate together, the core air the high pressure compressor and the high pressure turbine generating a high pressure first axial force on the spool; a high pressure balance piston that applies a high pressure second axial force on the high pressure spool to balance at least a portion of the high pressure first axial force, the high pressure balance piston comprising: a high pressure rotating boundary portion comprising a rotating portion of the high pressure spool; and a high pressure fixed enclosure portion that does not rotate, the high pressure rotating boundary portion and the high pressure fixed enclosure portion defining a high pressure balance piston

cavity, wherein the water system provides water to the high pressure balance piston cavity, the water in the high pressure balance piston cavity is a high pressure balance piston fluid, the high pressure balance piston fluid absorbs heat from the high pressure balance piston, causing an expansion and a pressure increase in the high pressure balance piston fluid, generating the high pressure second axial force.

4. The turbine engine of claim 1, wherein the water system further comprises a liquid water supply and an evaporator that converts the liquid water supply into a steam supply that is introduced into the balance piston.

5. The turbine engine of claim 4, the water system comprising a steam turbine, wherein the steam supply flows to the steam turbine, the steam turbine extracts a portion of thermal energy, kinetic energy, or thermal energy and kinetic energy from the steam supply, and the steam supply further flows to the combustor or to a piston cavity.

6. The turbine engine of claim 1, the shaft defining a secondary cavity, and the rotating boundary portion further comprising at least one passage therethrough, wherein the balance piston fluid flows through the passage and into the secondary cavity.

7. The turbine engine of claim 6, wherein the secondary cavity comprises turbine rotor disks that transfer heat to the balance piston fluid in the secondary cavity, to cool the turbine rotor disks.

8. The turbine engine of claim 6, wherein the at least one passage comprises a first passage, the shaft further comprises at least one second passage therethrough, and the balance piston fluid flows from the secondary cavity through the at least one second passage into the turbine.

9. The turbine engine of claim 6, wherein the secondary cavity is a plurality of secondary cavities, the at least one passage comprises a plurality of passages, and the plurality of passages connect to each of the plurality of secondary cavities allowing the balance piston fluid to flow through into each of the plurality of secondary cavities.

10. The turbine engine of claim 9, wherein the plurality of passages is a plurality of first passages, the shaft comprises a plurality of second passages therethrough, the plurality of second passages connecting to each of the plurality of secondary cavities, and the balance piston fluid flows from the secondary cavities through the respective second passages into the turbine.

11. A method of supporting axial force on the turbine engine of claim 1, the method comprising: providing water to the balance piston, the water becoming the balance piston fluid, transferring heat from the balance piston to the balance piston fluid to expand the balance piston fluid and to increase pressure of the balance piston fluid, and generating the second axial force.

12. The method of claim 11, wherein the water is in a liquid phase until the water enters the balance piston cavity, becoming two-phase steam or supercritical steam.

13. The method of claim 11, wherein the spool is a low pressure spool, the compressor is a low pressure compressor, the turbine is a low pressure turbine, the shaft is a low pressure shaft, the balance piston is a low pressure balance piston, the rotating boundary portion is a low pressure rotating boundary portion, the fixed enclosure portion is a low pressure fixed enclosure portion, the balance piston fluid is a low pressure balance piston fluid, the second axial force is a low pressure second axial force, and the turbine engine further comprises: a high pressure spool comprising: a high pressure compressor disposed between the low pressure compressor and the combustor that further compresses the core air from the low pressure compressor before the core air flows into the combustor; a high pressure turbine disposed between the combustor and the low pressure turbine, the high pressure turbine caused to rotate by the combustion gases prior to the combustion gases flowing into the low pressure turbine; and a high pressure shaft connecting the high pressure compressor and the high pressure turbine such that the high pressure compressor and the high pressure turbine rotate together, the core air, the high pressure compressor, and the high pressure turbine generating a high pressure first axial force on the spool; a high pressure balance piston that applies a high pressure second axial force on the high pressure spool to balance at least a portion of the high pressure first axial force, the high pressure balance piston comprising: a high pressure

rotating boundary portion comprising a rotating portion of the high pressure spool; and a high pressure fixed enclosure portion that does not rotate, the high pressure rotating boundary portion and the high pressure fixed enclosure portion defining a high pressure balance piston cavity, wherein the method further comprises providing water to the high pressure balance piston cavity, such that the water in the high pressure balance piston cavity, becoming a high pressure balance piston fluid, transferring heat from the high pressure balance piston to the high pressure balance piston fluid, expanding the high pressure balance piston fluid and increasing pressure of the high pressure balance piston fluid, and generating the high pressure second axial force.

14. The method of claim 11, wherein the water system comprises an evaporator, and the water is a liquid water supply prior to entering the evaporator, and the method further comprises converting the liquid water supply into a steam supply in the evaporator and introducing the steam supply into the balance piston.

15. The method of claim 14, wherein the water system further comprises a steam turbine, the method further comprising causing the steam supply to flow to the steam turbine, extracting a portion of thermal energy, kinetic energy, or thermal energy and kinetic energy from the steam supply.

16. The method of claim 11, wherein the shaft comprises a secondary cavity, and the rotating boundary portion further comprises at least one passage therethrough, the method further comprising causing the balance piston fluid to flow through the at least one passage and into the secondary cavity and transferring heat from the secondary cavity to the balance piston fluid.

17. The method of claim 16, wherein the secondary cavity comprises turbine rotor disks, the method further comprising transferring heat from the turbine rotor disks to the balance piston fluid in the secondary cavity, to cool the turbine rotor disks.

18. The method of claim 16, wherein the at least one passage comprises a first passage, and the shaft further comprises at least one second passage therethrough, and the method further comprises causing the balance piston fluid to flow from the secondary cavity through the at least one second passage into the turbine.

19. The method of claim 16, wherein the secondary cavity is a plurality of secondary cavities and the at least one passage comprises a plurality of passages, the plurality of passages connecting to each of the secondary cavities, the method further comprising causing the balance piston fluid to flow to the plurality of secondary cavities and transferring heat from each of the plurality of secondary cavities to the balance piston fluid in the respective secondary cavity.

20. The method of claim 19, wherein the plurality of passages is a plurality of first passages, the shaft further comprises a plurality of second passages therethrough, the plurality of second passages connecting to each of the plurality of secondary cavities, and the method further comprises causing the balance piston fluid to flow from each of the plurality of secondary cavities through the corresponding secondary passages into the turbine.
